
AZ104 Microsoft Azure Administrator

- The Azure Administrator implements, manages, and monitors identity, governance, storage, compute, and virtual networks in a cloud environment.
- The Azure Administrator provisions, sizes, monitors, and adjusts resources as appropriate.

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Birds Eye View

Think of AZ-104 as:

"Can you operate Azure day-to-day in production?"

Not architect it (AZ-305). Not develop for it (AZ-204). Not secure everything (AZ-500).

Just keep Azure running.

The Azure Administrator's Mental Model

Area	Common Admin Tasks	Primary Tools
Identity	Users, Groups, RBAC, Entra ID, MFA	Portal, Entra Admin Center
Governance	Policies, Tags, Locks, Management Groups	Portal, Azure Policy
Cost	Budgets, Cost Analysis, Reservations	Cost Management
Networking	VNets, Subnets, NSGs, DNS, Peering	Portal, Network Watcher
Storage	Blob, Files, Replication, SAS	Portal, Storage Explorer
Compute	VMs, Scale Sets, App Services	Portal, CLI, PowerShell
Monitoring	Metrics, Logs, Alerts	Azure Monitor
Backup/DR	Backup Vaults, ASR	Backup Center
Automation	Bicep, ARM, CLI, PowerShell	Cloud Shell

Based directly on the official learning paths and assessment guide.

What You Must Know Cold for the Exam

Identity & Governance

Task	Tool
Create user	Entra ID
Create dynamic groups	Entra ID
Assign roles	RBAC
Enforce standards	Azure Policy
Organize resources	Resource Groups
Organize subscriptions	Management Groups
Cost reporting	Tags + Cost Management

Exam favorites:

- Reader vs Contributor vs Owner
 - RBAC scope hierarchy
 - Policy vs RBAC
 - Tags
 - Management Groups
-

Networking

Task	Tool
Create VNet	Virtual Network
Secure traffic	NSG
Group workloads	ASG
Connect VNets	Peering
Route traffic	Route Tables
Private access to Storage/SQL	Private Endpoint
Name resolution	Azure DNS
Troubleshooting	Network Watcher

Exam favorites:

- NSG rules
 - Peering
 - Service Endpoint vs Private Endpoint
 - Route Tables
 - Load Balancer vs Application Gateway
-

Storage

Task	Tool
Store files	Azure Files
Store objects	Blob Storage
Temporary access	SAS
Replication	LRS/ZRS/GRS/GZRS
Lifecycle management	Blob Policies
File management	Storage Explorer

Exam favorites:

- Replication types
 - Hot/Cool/Archive tiers
 - SAS
 - Storage security
-

Compute

Task	Tool
Create VM	Portal/CLI
Secure VM access	Bastion
Scale VMs	VMSS
HA	Availability Zones
Host web apps	App Service

Containers	ACI / ACA
-------------------	-----------

Exam favorites:

- Availability Set vs Availability Zone
- Scale Up vs Scale Out
- VMSS
- App Service Plans

Monitor & Protect

Task	Tool
Collect metrics	Azure Monitor
Alerts	Action Groups
Query logs	Log Analytics
KQL queries	Kusto
Backups	Recovery Services Vault
DR	Azure Site Recovery

Exam favorites:

- Azure Monitor
- Metrics vs Logs
- Action Groups
- Backup vs Site Recovery

The Azure Admin Tool Progression

2019 Azure Admin

1 Portal

2022 Azure Admin

1 Portal
2 + PowerShell
3 + Azure CLI
4 + ARM Templates

2025-2026 Azure Admin

1 Portal
2 + Azure CLI
3 + PowerShell
4 + Bicep
5 + Azure Copilot

Bicep is now Microsoft's preferred Infrastructure-as-Code authoring experience over raw ARM JSON. Azure administrators are also expected to understand Azure Copilot workflows. [\[ppt notes | Txt\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

What AI Agents Change

This is probably the most important "future-proofing" discussion for the class.

Old World

Admin manually:

- Finds VM issues
 - Builds scripts
 - Checks NSGs
 - Reviews costs
 - Creates resources
 - Troubleshoots outages
-

Azure Copilot World

Admin asks:

"Why is VM01 slow?"

Copilot reviews telemetry.

Admin asks:

"Show all Storage Accounts open to the internet."

Copilot generates resource queries.

Admin asks:

"Create a Bicep template for a secure storage account."

Copilot writes it.

Admin asks:

"Identify my highest-cost resources."

Copilot analyzes spending.

Azure Copilot can generate Azure CLI, PowerShell, Terraform, Bicep, investigate resources, analyze costs, visualize networks, troubleshoot services, and increasingly perform actions through agent-based workflows. [\[learn.microsoft.com\]](https://learn.microsoft.com), [\[learn.microsoft.com\]](https://learn.microsoft.com), [\[azure.microsoft.com\]](https://azure.microsoft.com)

Does AI Replace the Azure Admin?

No.

The exam is still testing whether *you know what good looks like*.

AI can:

- ✓ Generate commands
- ✓ Generate Bicep
- ✓ Summarize resources
- ✓ Find problems
- ✓ Recommend fixes

But the human still decides:

- Which network design is correct
- Which permissions are appropriate
- Which DR strategy is required
- Which storage replication should be used
- Which compliance policy applies

Think:

AI is becoming the junior admin. AZ-104 teaches you how to be the senior admin.

The 30-Second AZ-104 Summary

If I had one slide before a 5-day bootcamp:

Domain	Question Microsoft Asks
Identity	Who can access Azure?
Governance	What rules control Azure?
Networking	How does traffic move?
Storage	Where is data stored?
Compute	Where do workloads run?
Monitoring	How do we know something broke?
Backup/DR	How do we recover when it breaks?
Automation	How do we do it repeatedly?

AI/Copilot	How do we do it faster?
------------	-------------------------

Memory aid:

People → Policy → Network → Storage → Compute → Monitor → Recover → Automate

That's AZ-104 in one sentence. 🚀

MS Entra ID Tenant Hive Forest

In **Microsoft Entra ID** (formerly Azure AD), **the traditional Active Directory concepts of forests and hives either do not exist or function entirely differently, while a tenant is the foundational building block of the cloud identity platform.**

Here is a direct comparison of how these three terms relate to Microsoft Entra ID:

Concept	Traditional Definition (On-Premises AD)	Microsoft Entra ID Point of View	Primary Purpose
Tenant	<i>Does not exist in traditional AD.</i>	The fundamental boundary of Entra ID. It represents a single organization or instance of the cloud service.	Houses all users, groups, and application registrations for an organization.
Forest	The highest security and administrative boundary containing domains, trees, and trust relationships.	Does not exist in Entra ID. Entra ID is flat, non-hierarchical, and has no domains or trusts within a single tenant.	Replaced by the Tenant boundary. (Multiple tenants require Entra ID Cross-Tenant Access).
Hive	A logical group of keys, subkeys, and values in the Windows Registry (e.g., HKEY_LOCAL_MACHINE).	Not an identity concept. It has no structural or architectural meaning in Entra ID.	Used purely for local Windows operating system configuration.

 Tenant: The Cloud Boundary

A tenant is a dedicated, isolated instance of Microsoft Entra ID that an organization receives when it signs up for a Microsoft cloud service (like Azure or Microsoft 365).

- **Scope:** It serves as the ultimate boundary for security, compliance, and administration.
- **Structure:** It is entirely flat. Unlike an on-premises forest, you cannot create "child domains" or organizational units (OUs) inside a tenant to partition users hierarchically.
- **Collaboration:** To connect two different tenants, you must use cloud-native features like **Entra ID B2B collaboration** or **Cross-Tenant Access Settings**, rather than traditional domain trusts.

 **Forest:** The On-Premises Legacy

In traditional on-premises Active Directory, a forest is the top-level container.

- **The Entra ID Shift:** Entra ID does not use the concept of a forest. If you have an on-premises forest, you project its identities into a cloud *tenant* using tools like **Microsoft Entra Connect Sync** or **Cloud Sync**.
- **Entra Domain Services exception:** If you use *Microsoft Entra Domain Services* (a managed service to run legacy domain-joined VMs in the cloud), you can create a managed domain that acts like a traditional forest, but this is a specialized PaaS feature, not the core Entra ID platform.

 **Hive:** A Local OS Component

The term hive is completely unrelated to Entra ID or identity management architecture.

- **The Windows Registry:** A hive is a registry database file used by the Windows operating system on local machines to store configuration data.
- **No Cloud Equivalent:** There is no cloud equivalent of a registry hive in Microsoft Entra ID, as Entra ID manages identity and access, not local operating system registry files.

URL Parts

URL part	Description
https://myaccount.blob.core.windows.net/container1/file1.pdf	URL to endpoint
?sv=2017-07-29	Rest API version
&st=2018-04-30T19%3A19%3A19Z	Validity start time
&se=2018-05-01T19%3A19%3A19Z	Validity end time
&sr=b	Type of resource
&sp=r	Permissions
&sip=168.1.5.60-168.1.5.70	IP Address / range (optional)
&spr=https	Protocol (optional)
&sig=pk9oGEPqYyu0K4Gutfreq9n0CJqgnjYgkEwcIEL8I0%3D	Signature

Azure AD authentication with RBAC

Exam Areas

Study Area	Percentage
Manage Azure identities and governance	20-25%
Implement and manage storage	15-20%
Deploy and manage Azure compute resources	20-25%
Configure and manage virtual networking	15-20%
Monitor and backup resources	10-15%

Training Path (5 Day)

This agenda is optimized for **AZ-104 exam success + real-world administrator skills**, using the official learning paths, assessment objectives, and the Microsoft Learning labs. The philosophy is:

Lecture only where concepts are difficult or tested heavily.

Everything else = learn by doing in labs.

The AZ-104 course is organized around Identity/Governance, Networking, Storage, Compute, and Monitoring/Protection domains, with 11 principal labs available. [\[learn.microsoft.com\]](https://learn.microsoft.com), [\[AZ-104T00A...smentGuide | PDF\]](#), [\[ppt_notes | Txt\]](#)

Day 1 Governance, Days 2-3 Networking, Day 4 Storage & Compute, Day 5 Monitoring & Protection mirrors both the exam blueprint and the way Azure administrators actually work.

Day 1

Azure Foundations, Identity & Governance

Morning

AZ-104 Overview

- Exam domains
- Azure admin vs architect vs engineer
- Azure resource hierarchy
 - Tenant
 - Management Groups
 - Subscription
 - Resource Groups
 - Resources

Administrator Tools

- Azure Portal
- Cloud Shell
- Azure CLI
- PowerShell
- Bicep
- Azure Copilot overview

Keep this section brief because students will use all tools throughout the week. [\[ppt_notes | Txt\]](#), [\[learn.microsoft.com\]](https://learn.microsoft.com)

Entra ID Administration

- Users
- Groups
- Dynamic Groups
- Administrative Roles
- Self-Service Password Reset
- Licensing concepts

Governance

- RBAC
- Scope hierarchy
- Reader / Contributor / Owner
- User Access Administrator
- Resource Locks
- Tags
- Azure Policy
- Management Groups

Exam emphasis:

- RBAC vs Policy
- Least privilege
- Scope inheritance

[\[AZ-104T00A...smentGuide | PDF\]](#), [\[ppt_notes | Txt\]](#)

Afternoon Labs

Lab 01

Manage Entra ID Identities

Students:

- Create users
 - Create groups
 - Assign users
 - Review roles
-

Lab 02a

Manage Subscriptions and RBAC

Students:

- Create role assignments
 - Test access
 - Understand scope inheritance
-

Lab 02b

Manage Governance via Azure Policy

Students:

- Tags
 - Policies
 - Resource Locks
 - Compliance
-

Day 1 Exam Takeaways

Know cold:

- Tenant hierarchy
 - RBAC roles
 - Policy effects
 - Tagging strategy
 - Management Groups
-

Day 2

Core Networking

This is traditionally the hardest AZ-104 section.

Morning Lecture

Virtual Networks

- Address spaces
 - Subnets
 - Private vs Public IP
 - CIDR basics
 - DNS concepts
-

Network Security

- NSGs
 - ASGs
 - Security rule processing
 - Service Tags
-

Azure DNS

- Public DNS
 - Private DNS
 - Name resolution
-

Afternoon Labs

Lab 04

Implement Virtual Networking

Students:

- Build VNets
 - Create subnets
 - Configure NSGs
 - Configure DNS
-

Discussion Block

Use Network Watcher Topology view.

Students draw:



This dramatically improves exam retention.

Day 2 Exam Takeaways

Know cold:

- NSG rules
 - ASG purpose
 - VNet design
 - DNS scenarios
 - IP addressing basics
-

Day 3

Hybrid Connectivity & Traffic Management

Build on Day 2.

Morning Lecture

VNet Peering

- Regional
- Global

Routing

- System Routes
- User Defined Routes (UDR)

Service Endpoints

Private Endpoints

Private Link

Most exam questions occur here.

Students must understand:

| Service Endpoint = secure path

vs

| Private Endpoint = private IP in your VNet

[\[AZ-104T00A...smentGuide | PDF\]](#), [\[ppt notes | Txt\]](#)

Load Balancing

- Azure Load Balancer
 - Internal vs Public
 - Application Gateway
 - Front Door overview
 - Traffic Manager overview
-

Network Troubleshooting

- Network Watcher
- IP Flow Verify
- Connection Troubleshoot
- NSG Diagnostics

Afternoon Labs

Lab 05

Implement Intersite Connectivity

Students:

- VNet Peering
 - Routes
 - Endpoints
-

Lab 06

Implement Network Traffic Management

Students:

- Load Balancer
 - Application Gateway
 - Path-based routing
 - Network diagnostics
-

Day 3 Exam Takeaways

Know cold:

- Peering requirements
 - UDR scenarios
 - Service Endpoint vs Private Endpoint
 - Load Balancer vs Application Gateway
 - Network Watcher tools
-

Day 4

Storage & Compute

This is the biggest "admin operations" day.

Morning Block A

Storage

- Storage Accounts

- Blob Storage
- Azure Files
- Replication

Important chart:

- | | |
|---|---------|
| 1 | LRS |
| 2 | ZRS |
| 3 | GRS |
| 4 | RA-GRS |
| 5 | GZRS |
| 6 | RA-GZRS |

Storage Security

- RBAC
 - SAS
 - Firewalls
 - Encryption
 - Lifecycle Management
-

Lab

Lab 07

Manage Azure Storage

Students:

- Storage Accounts
 - Blob Storage
 - File Shares
 - SAS
 - Security
-

Afternoon Block B

Virtual Machines

- VM sizing
- Availability Sets
- Availability Zones
- Bastion
- Disks

VM Scale Sets

- Scale Out
 - Scale In
-

App Services

- Plans
 - Scaling
 - Deployment Slots
 - Backups
-

Containers

Brief overview only

- ACI
 - ACA
 - AKS awareness
-

Labs

Lab 08

Manage Virtual Machines

Lab 09a

Implement Web Apps

(Optional time permitting)

Lab 09b

Azure Container Instances

Lab 09c

Azure Container Apps

Day 4 Exam Takeaways

Know cold:

- Availability Set vs Zone
 - Scale Up vs Scale Out
 - VMSS
 - App Service Plans
 - Storage replication
 - SAS
-

Day 5

Monitoring, Backup, Recovery & Exam Review

This day ties all previous topics together.

Morning Lecture

Azure Backup

- Recovery Services Vault
 - Backup policies
 - File share backups
 - VM backups
-

Site Recovery

Students only need:

- | |
|--------------------------------|
| 1 Backup = restore data |
| 2 |
| 3 Site Recovery = restore site |

That distinction appears frequently in exams. [\[AZ-104T00A...smentGuide | PDF\]](#)

Azure Monitor

- Metrics
 - Logs
 - Activity Log
 - Diagnostic Settings
-

Alerts

- Alert Rules
- Action Groups

- Alert States
-

Log Analytics

- Workspaces
- KQL basics

Example:

```
1 Event
2 | where EventLevelName == "Error"
```

[\[AZ-104T00A...smentGuide | PDF\]](#)

Afternoon Labs

Lab 10

Implement Data Protection

Students:

- Backup
 - Recovery Vault
 - Site Recovery
-

Lab 11

Implement Monitoring

Students:

- Monitor
 - Alerts
 - Action Groups
 - Log Analytics
 - KQL
-

Final 2-Hour AZ-104 Review Session

Rapid Fire Comparison Matrix

Topic	Know the Difference
RBAC vs Policy	Access vs Compliance
NSG vs ASG	Rules vs Grouping

Service Endpoint vs Private Endpoint	Secure path vs Private IP
Load Balancer vs App Gateway	L4 vs L7
Availability Set vs Zone	Datacenter vs Region resilience
Scale Up vs Scale Out	Bigger VM vs More VMs
Blob vs Files	Object vs SMB
Backup vs Site Recovery	Data vs Entire Site
Metrics vs Logs	Numbers vs Records

Recommended Daily Timing

Activity	Time
Lecture	40%
Guided Demo	15%
Labs	35%
Exam Review / Q&A	10%

For AZ-104 specifically, I'd target roughly **14-16 hours of lecture** and **18-20 hours of labs and exercises** across the five days. The Microsoft labs are strong enough that students often learn faster doing the task than watching slides. [\[learn.microsoft.com\]](https://learn.microsoft.com), [\[ppt notes | Txt\]](#)

Bottom line: Day 1 Governance, Days 2-3 Networking, Day 4 Storage & Compute, Day 5 Monitoring & Protection mirrors both the exam blueprint and the way Azure administrators actually work.

Updates to Content

Update – 13 February 2026

Before teaching be sure to review the slide decks.

- Awareness. There is new Learn content around AI-Ready Infrastructure. More content is expected, but here are two examples, Govern AI Ready Workloads with Microsoft Foundry and Manage monitoring for AI-Ready Infrastructure .

What Actually Retired

- **Azure Data Studio (ADS):** Retired on February 28, 2026. Microsoft recommends moving database management workloads to Visual Studio Code with the MSSQL extension.
- **Azure Resource Explorer:** Shutting down in favor of newer tooling like Azure Resource Graph.
- **Azure Storage Emulator:** Replaced and deprecated in favor of Azurite.

ARM Template – JSON

Azure Resource Manager Templates. Written in JavaScript. Used by Azure for almost everything that can be created and often added to DevOps Deployments.

Infrastructure-as-Code work is being authored in Bicep and compiled into ARM JSON before deployment.

ARM template engine is treating the parameters object like a lookup table (dictionary/hash map) and later "calling" values from it using the parameters() function. JSON stores the definition; ARM provides the execution layer that resolves those references.

JSON Structure & Parameters

Yes! You're noticing something fundamental about how ARM templates work, and it's a great observation.

The interesting thing is that **JSON itself isn't doing any variable substitution or "calling" anything**. JSON is just a data structure. The magic comes from the **ARM template engine**, which interprets certain strings as expressions.

Pure JSON View

If this were only JSON:

```
1 {  
2   "parameters": {  
3     "storageName": {  
4       "type": "string"  
5     }  
6   }  
7 }
```

JSON sees:

```
1 Object  
2   └─ parameters  
3     └─ storageName  
4         └─ type = string
```

Just nested name/value pairs.

No variables. No functions. No execution.

Where ARM Takes Over

Later you see:

```
1 "name": "[parameters('storageName')]"
```

To JSON, that's just a string:

```
1 "[parameters('storageName')]"
```

Nothing more.

But when ARM deploys the template, it sees the square brackets:

```
1 [ ... ]
```

and says:

```
"Ah, that's an ARM expression."
```

Now ARM executes:

```
1 parameters('storageName')
```

which means:

```
Look in the parameters collection and retrieve the value supplied for storageName.
```

Conceptually:

```
1 Parameters Collection
2
3 storageName = mystorageacct01
```

ARM evaluates:

```
1 parameters('storageName')
```

and returns:

```
1 mystorageacct01
```

So:

```
1 "name": "[parameters('storageName')]"
```

becomes

```
1 "name": "mystorageacct01"
```

during deployment.

It's Almost Like a Dictionary Lookup

If you're a Python guy, think:

```
1 parameters = {
2     "storageName": "mystorageacct01"
3 }
4
5 parameters["storageName"]
```

returns:

```
1 "mystorageacct01"
```

ARM's:

```
1 parameters('storageName')
```

is essentially doing:

```
1 parameters["storageName"]
```

behind the scenes.

Why It Feels Like an Array

Many people initially think:

```
1 "parameters": {
```

is an array.

But it isn't.

Array:

```
1 [  
2   "one",  
3   "two",  
4   "three"  
5 ]
```

Object:

```
1 {  
2   "one": 1,  
3   "two": 2,  
4   "three": 3  
5 }
```

Your ARM parameter block is actually a nested object:

```
1 {  
2   "parameters": {  
3     "storageName": {  
4       ...  
5     }  
6   }  
7 }
```

Think:

```
1 parameters  
2   └─ storageName  
3       └─ metadata
```

A hierarchy of dictionaries/hash tables.

ARM Template Mental Model

I teach students to think of ARM templates as:

```
1 1. Parameters
```

```
2     INPUTS
3
4  2. Variables
5     CALCULATIONS
6
7  3. Resources
8     DEPLOYMENTS
9
10  4. Outputs
11     RESULTS
```

Like this:

```
1  User Input
2      ↓
3  Parameters
4      ↓
5  Variables
6      ↓
7  Resources
8      ↓
9  Outputs
```

Example Walkthrough

Suppose a user launches the deployment and enters:

```
1  {
2    "storageName": "timstorage001"
3  }
```

ARM stores:

```
1  storageName = timstorage001
```

Then reaches:

```
1  "name": "[parameters('storageName')]"
```

and evaluates:

```
1  parameters('storageName')
```

Result:

```
1  timstorage001
```

Final resource sent to Azure:

```
1  {
2    "type": "Microsoft.Storage/storageAccounts",
3    "name": "timstorage001"
```

```
4 }
```

Why Bicep Feels Easier

This is exactly why Microsoft pushes Bicep now.

ARM:

```
1 "name": "[parameters('storageName')]"
```

Bicep:

```
1 param storageName string
2
3 resource stg 'Microsoft.Storage/storageAccounts@2025-01-01' = {
4   name: storageName
5 }
```

The Bicep version looks much more like a modern programming language.

Under the covers, however:

```
1 Bicep
2   ↓ compile
3   ARM JSON
4   ↓ deploy
5   Azure Resource Manager
```

So AZ-104 students should understand ARM syntax, but in the real world most new Infrastructure-as-Code work is being authored in Bicep and compiled into ARM JSON before deployment.

Your insight is exactly right: the ARM template engine is treating the parameters object like a lookup table (dictionary/hash map) and later "calling" values from it using the parameters() function. JSON stores the definition; ARM provides the execution layer that resolves those references.

User Tasks/Exercises

1. [Get started with Azure Cloud Shell using persistent storage - Azure Cloud Shell | Microsoft Learn](#)
2. [Tutorial: Build policies to enforce compliance - Azure Policy | Microsoft Learn](#)
3. [Exercise - Create and deploy an Azure Resource Manager template - Training | Microsoft Learn](#)
4. [Exercise - Add parameters and outputs to your Azure Resource Manager template - Training | Microsoft Learn](#)
5. [How to manage groups - Microsoft Entra | Microsoft Learn](#)
6. [Create a Microsoft Customer Agreement subscription - Azure Cost Management + Billing | Microsoft Learn](#)
7. [Tutorial: Build policies to enforce compliance - Azure Policy | Microsoft Learn](#)
8. [Quickstart - Check access for a user to a single Azure resource - Azure RBAC | Microsoft Learn](#)
9. [Tutorial: Grant a user access to Azure resources using the Azure portal - Azure RBAC | Microsoft Learn](#)
10. [Analyze, estimate and optimize cloud costs using Azure Copilot | Microsoft Learn](#)

11. [Generate PowerShell scripts using Azure Copilot | Microsoft Learn](#)
12. [Manage Azure portal settings and preferences - Azure portal | Microsoft Learn](#)
13. [Create a dashboard in the Azure portal - Azure portal | Microsoft Learn](#)
14. [How to create an Azure support request - Azure portal | Microsoft Learn](#)
15. [Tutorial - Create and deploy template - Azure Resource Manager | Microsoft Learn](#)
16. [Quickstart: Create an Azure Virtual Network | Microsoft Learn](#)
17. [Tutorial: Restrict access to PaaS resources with service endpoints | Microsoft Learn](#)
18. [Tutorial: Host your domain in Azure DNS | Microsoft Learn](#)
19. [Tutorial: Create an alias record to refer to a resource record in a zone - Azure DNS | Microsoft Learn](#)
20. [Tutorial: Connect virtual networks with peering | Microsoft Learn](#)
21. [Tutorial: Route network traffic with a route table | Microsoft Learn](#)
22. [Quickstart: Create a public load balancer - Azure portal - Azure Load Balancer | Microsoft Learn](#)
23. [Quickstart: Direct web traffic using the portal - Azure Application Gateway | Microsoft Learn](#)
24. [Create an Azure Storage Account - Azure Storage | Microsoft Learn](#)
25. [Quickstart: Upload, download, and list blobs - Azure portal - Azure Storage | Microsoft Learn](#)
26. [Create shared access signature \(SAS\) tokens for your storage containers and blobs - Foundry Tools | Microsoft Learn](#)
27. [Create a classic file share - Azure Files | Microsoft Learn](#)
28. [Get started with Storage Explorer | Microsoft Learn](#)
29. [Copy or move data to Azure Storage by using AzCopy v10 | Microsoft Learn](#)
30. [Quickstart - Create a Windows VM in the Azure portal - Azure Virtual Machines | Microsoft Learn](#)
31. [Quickstart - Create a Linux VM in the Azure portal - Azure Virtual Machines | Microsoft Learn](#)
32. [Quickstart: Deploy Azure Bastion from the Azure portal - Azure Bastion | Microsoft Learn](#)
33. [Create virtual machines in a Flexible scale set using Azure portal - Azure Virtual Machine Scale Sets | Microsoft Learn](#)
34. [Manage an App Service Plan - Azure App Service | Microsoft Learn](#)
35. [Scale Up Features and Capacities - Azure App Service | Microsoft Learn](#)
36. [How to Enable Automatic Scaling - Azure App Service | Microsoft Learn](#)
37. [Exercise - Create a web app in the Azure portal - Training | Microsoft Learn](#)
38. [Quickstart - Deploy Docker container to container instance - Portal - Azure Container Instances | Microsoft Learn](#)
39. [Quickstart: Deploy Your First Container App Using the Azure portal | Microsoft Learn](#)
40. [Back up Azure Files in the Azure portal - Azure Backup | Microsoft Learn](#)
41. [Tutorial - Back Up Multiple Azure Virtual Machines by Using Azure Backup - Azure Backup | Microsoft Learn](#)
42. [Create Azure Monitor metric alert rules - Azure Monitor | Microsoft Learn](#)
43. [Log Analytics Tutorial: Query and Analyze Log Data - Azure Monitor | Microsoft Learn](#)

Reference

1. [Microsoft Certified: Azure Administrator Associate - Certifications | Microsoft Learn](#)
2. [Course AZ-104T00-A: Microsoft Azure Administrator - Training | Microsoft Learn](#)

3. [MicrosoftLearning/AZ-104-MicrosoftAzureAdministrator: AZ-104 Microsoft Azure Administrator](#)
4. Videos: [Preparing for AZ-104 - Manage Azure identities and governance \(1 of 5\) | Microsoft Learn](#)
5. Sandbox Exam: [Microsoft Certification Exam Candidate Agreement - AA-001](#)
6. [Study guide for Exam AZ-104: Microsoft Azure Administrator | Microsoft Learn](#)
7. End list

Notes that make other notes and will get deleted?

MC Azure Administrator Study Guide: Exam AZ-104 Syllabus outline

Course Description

This syllabus is intended to act as a guide and starting point to instructors who are using the AZ-104 Microsoft Azure Administrator Study Guide as the text to support students taking the AZ-104 exam. It is also meant to be a companion to the slides included in the instructor material.

The book content has been designed to help individuals with fundamental knowledge of Microsoft Azure to learn about Azure administration for taking and passing the certification exam. Some students have a strong background of core Azure services, Azure workloads, security, and governance. Other students might have a little or zero technical expertise in Azure. Whatever the case, the content within this book is appropriate for a broad spectrum of learners and serves as the foundation for any student who wishes to become a subject matter expert in any of the covered topics. Having said that, if your students lack fundamental knowledge of Azure cloud, feel free to refer to the Microsoft documentation and learning paths provided by Microsoft.

Since publication chances are there some features and portal appearances might have been changed. Feel free to refer to Microsoft documentation for latest updates and changes. Also, chances are there that the exam syllabus also gets updated, at the time of writing this book, the last exam refresh was in September 2021. I would recommend checking the syllabus from time to time to understand if there is addition or omission of topics. Nevertheless, the core topics covered in this book are very unlikely to change and will help to build a foundation.

Before beginning the class, students should have basic understanding of Azure platform and terminologies. While not necessary, basic knowledge of PowerShell, Windows, and Linux will certainly help.

General Class Format

AZ-104 content is a mix of conceptual and hands-on experience. For example, we will start with the explanation, benefits and use case scenarios of a service, then we will go ahead with implementation of the service. Consider having an open discussion about scenarios, benefits, challenges, etc. about the topics covered in the session to go along with the hands-on lab.

A typical class will be sixty to ninety minutes long, broken down roughly as follows:

Ten minutes — Review of previous lecture's material

Fifty minutes — Lecture

Ten minutes — Review and questions

Twenty minutes — Lab, scenario discussions, and exam tips

Some chapters of the book consume multiple sessions, while other chapters consume fewer where there is less material to be covered. Thus, instead of sticking with a full twenty minutes of lab time each session where it might not be as valuable, you could place a longer lab in the final session for each chapter and adjust the lecture time to accommodate it. This will follow naturally depending on the capabilities of each class as you may find you need to spend a bit longer recapping some of the more challenging concepts.

Of course, you may have your own preferred structure, and some days will require changes.

Course Schedule

Session 1

Introductions and course overview. Be sure to take some time to get to know your students. Pay attention to their relative skill levels—ask them to introduce themselves and give a bit of background about their experience using not just Azure cloud, but other topics like Windows, Linux, and PowerShell. Knowing how technical the students in the room are will help you adjust your sessions and style to ensure that all students have the opportunity to be successful.

Chapter 1, “Identity - Azure Active Directory”

Text sections: “Azure Active Directory”, “User and Groups”,

Session 2

Chapter 1, “Identity - Azure Active Directory” (continued)

Text sections: “Azure AD Roles”, “Azure AD Join”

Session 3

Chapter 1, “Identity - Azure Active Directory” (continued)

Text sections: “Self-service Password Reset”, “Managing Multiple Directories”, Chapter Summary, and Review Questions

Session 4

Chapter 2, “Compliance and Cloud Governance”

Text sections: “Azure Regions”, “Azure Accounts and Subscriptions”, “Azure Cost Management”,

Session 5

Chapter 2, “Compliance and Cloud Governance” (continued)

Text sections: “Resource Groups”, “Management groups”, “Azure Policy”

Session 6

Chapter 2, "Compliance and Cloud Governance" (continued)

Text sections: "Role-based access control"

Session 7

Chapter 2, "Compliance and Cloud Governance" (continued)

Text sections: "Resource Locks", "Resource Tags", Chapter Summary, and Review Questions

Session 8

Chapter 3, "Virtual Networking"

Text sections: "VNet Concepts", "IP Addressing", "Network Routes",

Session 9

Chapter 3, "Virtual Networking" (continued)

Text sections: "Service Endpoints", "Private Endpoints"

Session 10

Chapter 3, "Virtual Networking" (continued)

Text sections: "Azure DNS", "Network Security Groups"

Session 11

Chapter 3, "Virtual Networking" (continued)

Text sections: "Azure Firewall", Chapter Summary, and Review questions

Session 12

Chapter 4, "Intersite Connectivity"

Text sections: "Azure-to-Azure connectivity"

Session 13

Chapter 4, "Intersite Connectivity" (continued)

Text sections: "Azure to on-premises connectivity",

Session 14

Chapter 4, "Intersite Connectivity" (continued)

Text sections: "Intersite Connectivity Architecture"

Session 15

Chapter 4, "Intersite Connectivity" (continued)

Text sections: "Virtual WAN", Chapter Summary, and Review questions

Session 16

Chapter 5, "Network Traffic Management"

Text sections: "Availability options"

Session 17

Chapter 5, "Network Traffic Management" (continued)

Text sections: "Azure Load Balancer"

Session 18

Chapter 5, "Network Traffic Management" (continued)

Text sections: "Azure Application Gateway"

Session 19

Chapter 5, "Network Traffic Management" (continued)

Text sections: "Azure Front Door", "Azure Traffic Manager", "Comparing the load balancing solutions", Chapter Summary, and Review Questions.

Session 20

Chapter 5, "Azure Storage"

Text sections: "Azure Storage Account", "Azure Storage Services", "Storage Replication"

Session 21

Chapter 5, "Azure Storage" (continued)

Text sections: "Storage account endpoint" and "Azure Blob Storage"

Session 22

Chapter 6, "Azure Storage" (continued)

Text sections: "Storage security", "Azure Files and File Sync"

Session 23

Chapter 6, "Azure Storage" (continued)

Text sections: "Managing Storage", Chapter Summary and Review Questions

Session 24

Chapter 7, "Azure Virtual Machines"

Text sections: "Virtual Machine Planning"

Session 25

Chapter 7, "Azure Virtual Machines" (continued)

Text sections: "Connecting to Virtual Machines"

Session 26

Chapter 7, "Azure Virtual Machines" (continued)

Text sections: "Scaling Concepts", "Virtual Machine Scale Set", Chapter Summary, and Review Questions.

Session 27

Chapter 8, "Automation, Deployment, and Configuration of Resources"

Text sections: "ARM templates"

Session 28

Chapter 8, "Automation, Deployment, and Configuration of Resources" (continued)

Text sections: "Configuring Virtual Hard Disk", "Virtual Machine Extensions", Chapter Summary and Review Questions

Session 29

Chapter 9, "PaaS Compute Options"

Text sections: "Azure App Service Plans", "Azure App Services"

Session 30

Chapter 9, "PaaS Compute Options" (continued)

Text sections: "Azure App Services" (continued), "Container Instances"

Session 31

Chapter 9, "PaaS Compute Options" (continued)

Text sections: "Azure Kubernetes Services", Chapter Summary and Review Questions

Session 32

Chapter 10, "Data Protection"

Text sections: "Files and Folder Backup", "Virtual Machine Data Protection"

Session 33

Chapter 10, "Data Protection" (continued)

Text sections: "Virtual Machine Data Protection" (continued), Chapter Summary and Review Questions

Session 34

Chapter 11, "Monitoring Resources"

Text sections: "Azure Monitor"

Session 35

Chapter 11, "Monitoring Resources" (continued)

Text sections: "Azure Alerts"

Session 36

Chapter 11, "Monitoring Resources" (continued)

Text sections: "Log Analytics"

Session 37

Chapter 11, "Monitoring Resources" (continued)

Text sections: "Network Watcher", Chapter Summary, and Review Questions

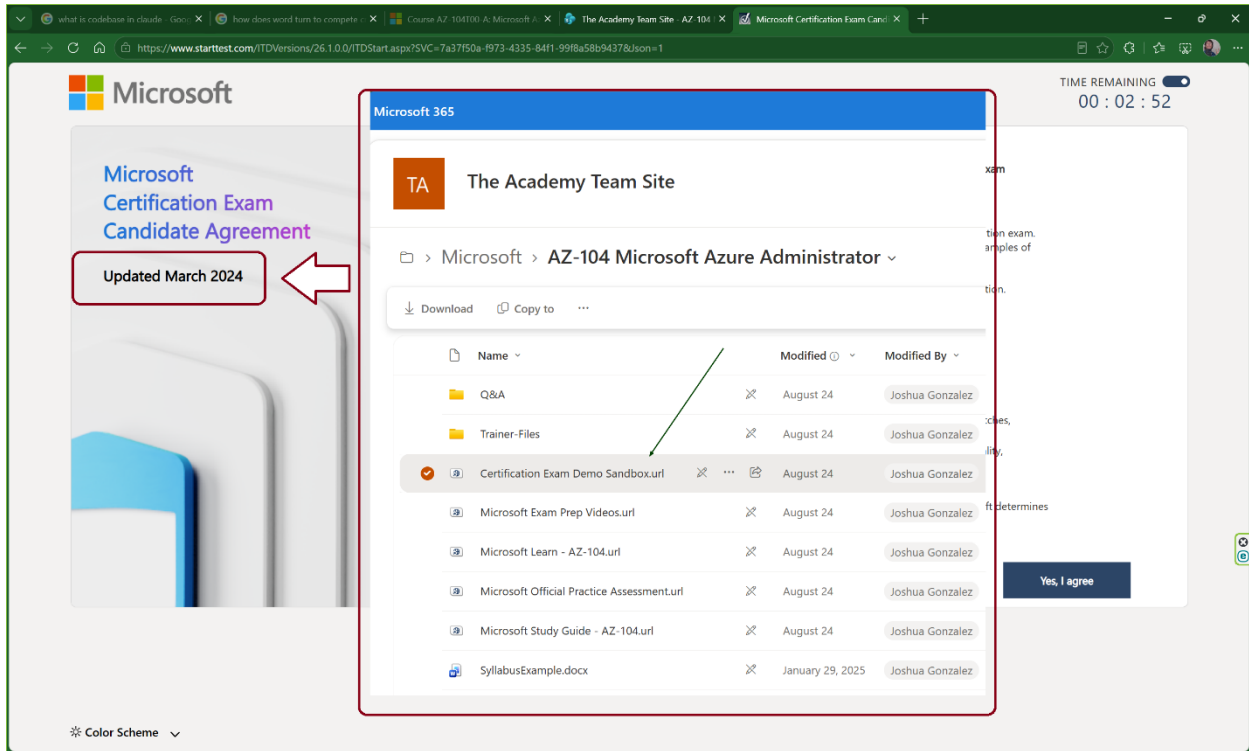
Session 38

Recap on materials and Q&A from students about any previous topic.

Extracted Notes

Other areas under updates

Exam Sandbox



Microsoft

Microsoft Certification Exam Candidate Agreement

Updated March 2024

Microsoft 365

TA The Academy Team Site

> Microsoft > AZ-104 Microsoft Azure Administrator

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Name	Modified	Modified By
Q&A	August 24	Joshua Gonzalez
Trainer-Files	August 24	Joshua Gonzalez
☒ Certification Exam Demo Sandbox.url	August 24	Joshua Gonzalez
☐ Microsoft Exam Prep Videos.url	August 24	Joshua Gonzalez
☐ Microsoft Learn - AZ-104.url	August 24	Joshua Gonzalez
☐ Microsoft Official Practice Assessment.url	August 24	Joshua Gonzalez
☐ Microsoft Study Guide - AZ-104.url	August 24	Joshua Gonzalez
☐ SyllabusExample.docx	January 29, 2025	Joshua Gonzalez

Yes, I agree

Color Scheme